

Interaction and Moderation

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Chapter overview

An additive straight-line model gives each predictor one partial slope. This chapter asks when a comparison should instead vary across another predictor's values or groups. A product term makes that possibility explicit. We follow its meaning from generated practice-and-preparation records to faculty salaries and conformity responses: interpret the coefficients at their anchors, estimate conditional comparisons, show where the records support them, and evaluate their uncertainty. By the end, you can distinguish a varying association from a causal explanation and write an informative report even when the slopes are poorly distinguished.

The same equation can pose different research questions. An extra hour of practice may have different associations with achievement at different preparation levels. A year's service may have different salary associations across academic ranks. We begin with the comparison that an additive model holds fixed, then examine what changes when it is allowed to vary.

1 A Comparison That Depends on Another Predictor

1.1 The restriction the additive model imposes

Suppose a researcher asks whether students who spend more time in deliberate practice tend to achieve higher posttest scores. An additive regression can allow students with stronger prior preparation to have higher scores while requiring the practice-score relationship to be the same at every preparation level. That requirement may be the very claim the researcher needs to examine. Perhaps an additional hour of practice is associated with a small score difference for students with lower prior preparation and a larger difference for students with higher prior preparation. Or perhaps the apparent difference is only sampling noise.

The central question is therefore conditional:

How does the expected outcome difference associated with one unit of a focal predictor vary across meaningful values or groups of another predictor?

An **interaction** is the model term that permits such variation. On the modeled scale, interaction means departure from additivity. Parallel lines are additive; nonparallel lines are interactive. The lines need not cross inside the observed data, and a statistically detectable product term does not by itself identify a causal mechanism.

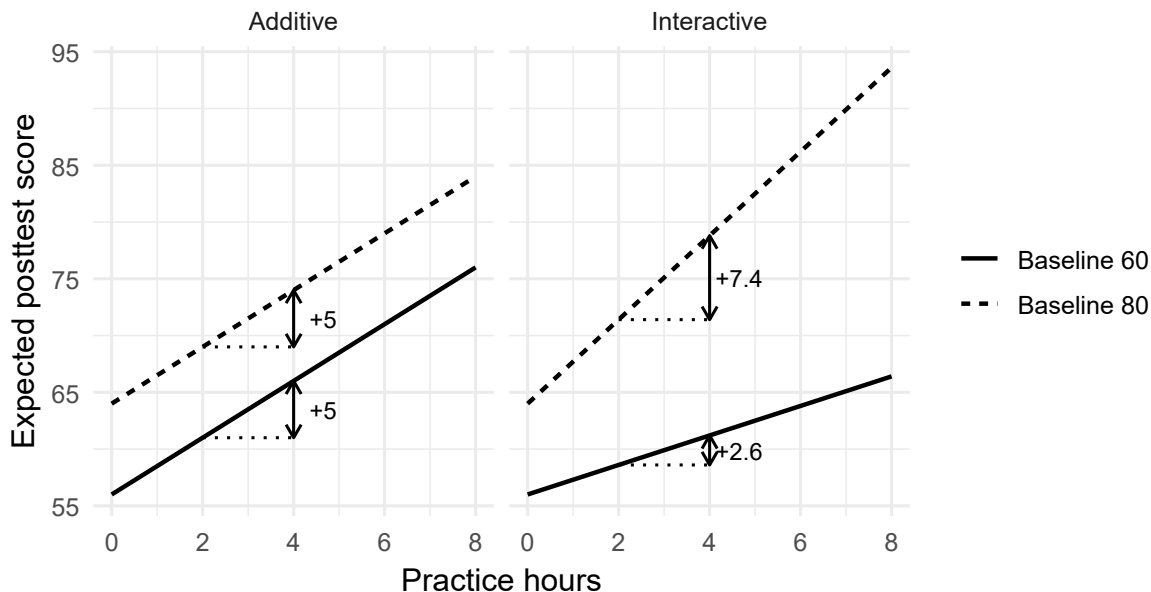


Figure 1: The same practice model with and without its product term. Arrows compare two with four practice hours at baseline 60 and 80: both differences are 5 under additivity, versus 2.6 and 7.4 under interaction. These are conditional model means.

Read the arrows as conditional comparisons. In the interactive panel, two additional practice hours correspond to 2.6 score points at baseline 60 and 7.4 at baseline 80. Their difference is 4.8 points. The product term will express this changing comparison in one equation. The lines are model means at two named baseline values: they are neither individual students' trajectories nor separate regressions fitted after splitting the students into groups. Section 2.4 declares this model and generates one sample from it.

This chapter first uses a simulation in which the conditional relationship is known. It then studies a real observational example: nine-month salary, years of service, and academic rank for the faculty of one U.S. college in 2008-09, from the package-distributed `carData::Salaries` data. The two examples play different roles. The simulation verifies algebra and repeated-sample logic against known truth. The salary analysis forces the harder decisions about units, severely unequal support, factor coding, uncertainty, diagnostics, and bounded reporting — and it shows what an honest interaction report looks like when the data do not clearly distinguish rank-specific slopes from one common slope.

1.2 The same idea for groups and continuous predictors

Chapter 4's factorial model allowed the practice-schedule difference to vary across feedback groups: its interaction test gave $F(2, 18) = 9.6$, $p = .0015$, and its declared difference of differences was 4 with SE 1.8257. Chapter 6 fitted that model in R. Here a continuous predictor replaces one factor, so a surface replaces a finite cell table and a moving slope replaces a single difference of differences. The ANOVA foundations remain the same; what changes is the form of the conditional comparison.

1.3 Interaction is defined on a stated scale

Additivity is always defined on a scale. Lines that are nonparallel for salary in dollars remain nonparallel after converting dollars to thousands of dollars, although every outcome coefficient is divided by 1,000. A nonlinear transformation such as a logarithm can change whether a pattern appears additive because it changes the model itself, not merely the label on the axis.

The same caution matters in later generalized linear models: interaction on a linear-predictor scale need not look the same on a response scale. This chapter stays with continuous outcomes and the identity-link linear model. Logistic interaction, multilevel interaction, and three-way interaction are outside its worked scope.

Moderation is not mediation or mechanism

A moderator names where a focal relationship is allowed to vary. That reading describes when an association differs, rather than how an effect is transmitted. An intervening variable and a causal mechanism need additional design and identification evidence. Chapter 14 develops those requirements.

2 The Product Term and Its Coefficients

2.1 The product coefficient is a difference in slopes

Begin with two continuous predictors, X and W , and a continuous outcome, Y . An additive conditional-mean model is

$$E(Y_i | X_i, W_i) = \beta_0 + \beta_X X_i + \beta_W W_i.$$

Holding W fixed, the expected difference associated with a one-unit difference in X is always β_X . Holding X fixed, the expected difference associated with a one-unit difference in W is always β_W . The model permits different levels but not different slopes.

The product-term model is

$$E(Y_i | X_i, W_i) = \beta_0 + \beta_X X_i + \beta_W W_i + \beta_{XW} X_i W_i. \quad (1)$$

Regrouping the terms that involve X gives

$$E(Y_i | X_i, W_i) = (\beta_0 + \beta_W W_i) + (\beta_X + \beta_{XW} W_i) X_i.$$

The conditional slope of X at $W = w$ is therefore

$$\frac{\partial E(Y | X, W)}{\partial X} = \beta_X + \beta_{XW} w. \quad (2)$$

If $\beta_{XW} = 0$, the slope does not vary with W and the model is additive. If $\beta_{XW} \neq 0$, a one-unit difference in W changes the conditional slope of X by β_{XW} . This is what the product coefficient measures. It is a **difference in slopes**, not the slope of X at every value of W .

The same model can be regrouped around W :

$$\frac{\partial E(Y | X, W)}{\partial W} = \beta_W + \beta_{XW} x. \quad (3)$$

The algebra is symmetric: each predictor's conditional slope depends on the other. Substantive language need not be symmetric. Researchers often call W a **moderator** when theory makes X the focal predictor and W the variable across which the X relationship is expected to vary; the conditional slope $\beta_X + \beta_{XW} w$ is then called a **simple slope** or conditional effect at w (Aiken & West, 1991; Hayes, 2022). That label comes from the research question, not from a privileged side of the multiplication sign.

For two values x_a and x_b at the same moderator value w , the conditional mean contrast is

$$E(Y | x_b, w) - E(Y | x_a, w) = (x_b - x_a) (\beta_X + \beta_{XW}w).$$

That sentence contains several safeguards:

- it identifies the outcome scale;
- it identifies the focal predictor difference;
- it holds the other predictor at a stated value;
- it describes a conditional mean, not every case; and
- it remains associational unless the research design supports a causal interpretation.

2.2 Hierarchy: the lower-order terms stay

The formula

```
y ~ x * w
```

expands to $x + w + x:w$. The lower-order terms belong with their product term. This **hierarchy principle** has both mathematical and interpretive reasons (Cohen et al., 2003; Kutner et al., 2005).

First, deleting X while retaining XW constrains the slope of X to zero when $W = 0$. Deleting W imposes the analogous constraint at $X = 0$. Such constraints are rarely part of the research question. Second, shifting either predictor changes which lower-order coefficients represent slopes and differences. A hierarchical model remains the same fitted surface after a change of origin; a product-only model generally does not. Third, factor product terms are often represented by several columns, so retaining the full lower-order structure keeps the block's meaning coherent.

Hierarchy does not mean every possible product must be added. The interaction should be motivated by the research question, graph, design, or a prespecified model comparison. It means that once a product is in the model, its component terms remain.

2.3 Centering changes the question at zero

Let $W_c = W - c$. Substituting $W = W_c + c$ into Equation 1 gives

$$E(Y | X, W_c) = (\beta_0 + c\beta_W) + (\beta_X + c\beta_{XW})X + \beta_W W_c + \beta_{XW} X W_c.$$

Centering W at c therefore changes the intercept and the lower-order coefficient of X . The new X coefficient is the conditional slope at the original value $W = c$. It does not change the interaction coefficient, fitted values, residuals, or the corresponding full-versus-reduced

product-block test. Centering is a **reparameterization**: it puts zero at a value that makes a coefficient worth interpreting (Aiken & West, 1991).

Centering is not a remedy for confounding, selection bias, measurement error, omitted non-linearity, lack of overlap, or a poorly specified research question. It can change correlations among constructed columns and may make numerical output easier to read, but it does not add information or make unsupported comparisons valid.

Scaling is another reparameterization. If $X_{10} = X/10$, then a one-unit difference in X_{10} is a ten-unit difference in X . Coefficients and standard errors change units together; fitted values and substantive conclusions should agree after correct translation.

2.4 A known-truth example: what one sample recovers

Consider a deliberately simplified educational setting. Each row is one student. The focal predictor is weekly deliberate practice in hours, X . The second predictor is a baseline assessment score, W . The outcome is a posttest score, Y . The simulation asks an associational model question under known conditions: how does the practice-score slope vary with baseline score?

The data-generating process centers baseline score at 70:

$$\begin{aligned} Y_i &= 60 + 2.5X_i + 0.4(W_i - 70) \\ &\quad + 0.12X_i(W_i - 70) + \varepsilon_i, \\ \varepsilon_i &\sim \mathcal{N}(0, 5^2). \end{aligned} \tag{4}$$

The coefficients are fixed by design. At a baseline score of 70, the true practice slope is 2.5 score points per hour. Ten baseline points below that anchor, it is $2.5 + 0.12(-10) = 1.3$ points per hour. Ten points above, it is $2.5 + 0.12(10) = 3.7$ points per hour. The positive product coefficient does not say that practice changes baseline preparation, or vice versa; it says that the conditional mean surface is nonadditive.

Practice hours and baseline scores are generated independently here so that the interaction mechanics are transparent. Real educational data would demand attention to attendance, access to support, prior courses, measurement reliability, clustering within classrooms or schools, and the process by which practice time was chosen. None of those design complications is solved by fitting a product term.

The explicit seed makes this one sample reproducible. It does not make the sample representative of a real population. The observed baseline range is 42.8747 to 96, and practice lies between 0.0141 and 7.9802 hours. Conditional estimates outside this rectangle would be extrapolations even though the generating equation is known to us.

The working object `sim` holds `score`, `hours`, `base` and `baseline_c = base - 70`. Generating the baseline, practice and independent error in that order with seed 909 gives the declared sample.

```
set.seed(909)
base <- pmin(pmax(rnorm(320, 70, 10), 42), 96)
hours <- runif(320, 0, 8)
baseline_c <- base - 70
score <- 60 + 2.5*hours + .4*baseline_c +
  .12*hours*baseline_c + rnorm(320, 0, 5)
sim <- data.frame(score, hours, base, baseline_c)
m1 <- lm(score ~ hours * baseline_c, data = sim)
```

The R formula uses `*` to include both lower-order terms and their product. The table below reports each estimate with its standard error (SE), the estimated standard deviation of that estimate across hypothetical repeated samples, and a 95% confidence interval.

Table 1

One generated sample against its declared targets (seed 909, 320 rows).

Term	Target	Estimate	SE	95% lower	95% upper
Intercept	60.00	59.6123	0.5550	58.5204	60.7043
Practice	2.50	2.6972	0.1244	2.4524	2.9421
Centered baseline	0.40	0.3822	0.0625	0.2593	0.5051
Practice by centered baseline	0.12	0.1149	0.0139	0.0875	0.1422

One noisy sample does not reproduce every truth exactly. The fitted practice slope at baseline 70 is 2.6972 points per hour, compared with the true value 2.5. The fitted interaction is 0.1149 points per hour per baseline-score point, compared with the true value 0.12. Its interval is [0.0875, 0.1422]. The interval quantifies repeated-sample uncertainty under the fitted model; it does not quantify every possible source of design or measurement error.

The fitted lower-order coefficient for centered baseline, 0.3822, is the baseline score slope when practice is zero. It is not a baseline-score slope averaged over all practice values. In an interaction model, every lower-order coefficient is conditional on the other component being at its coded zero or reference level.

2.5 The product block against the additive restriction

The reduced model is written first because it states the restriction being evaluated: one common practice slope and one common baseline slope. The full model releases the common-slope restriction by adding the product.

The nested comparison gives

$$F = \frac{(RSS_R - RSS_F)/q}{RSS_F/(n - p)} = \frac{(9108.7071 - 7488.7866)/1}{7488.7866/316} = 68.3548, \quad p < .0001.$$

Here $q = 1$ added coefficient and $p = 4$ total coefficients. The product coefficient's $t = 8.2677$ has the same squared statistic from full precision. Both models use the same 320 outcomes.

Here the product block has one degree of freedom because both predictors are continuous. The comparison asks whether allowing the practice slope to vary linearly with baseline score reduces residual variation beyond the additive model. It does not test whether the entire model is true, whether the relationship is causal, or whether the interaction is large enough to matter in a particular decision.

In the one-degree-of-freedom case, the block F statistic equals the squared t statistic for the product coefficient. That identity does not generalize to a factor with three levels: a continuous-by-three-level factor interaction requires two product columns and a joint two-degree-of-freedom test.

```
m0 <- lm(score ~ hours + baseline_c, data = sim)
anova(m0, m1)
```

2.6 Recentering leaves the surface alone

The centered model uses baseline 70 as zero. A raw-score model uses the unobserved and impossible baseline score zero as its anchor.

Table 2

What centering changes and what it leaves alone.

Quantity	Raw-score model	Centered model
Coefficient of practice at the coded zero	−5.3448	2.6972
Lower-order baseline coefficient	0.3822	0.3822
Product coefficient	0.1149	0.1149
Product block F	68.3548	68.3548
Maximum fitted-value difference	below 10^{-12}	below 10^{-12}

The raw model's practice coefficient is the fitted practice slope at a baseline score of zero. Its unattractive value is not evidence that the model failed; it answers an unsupported question. After centering, the practice coefficient answers the supported question at baseline 70. The product coefficient and every fitted value agree to numerical precision.

Centering can change the p-values of lower-order coefficients because it changes the hypotheses they test. It should not change the one-degree- of-freedom product test in this corresponding model. A lower-order p-value becoming smaller after centering is not repaired confounding; it is a different conditional comparison.

```
raw <- lm(score ~ hours * base, data = sim)
max(abs(fitted(raw) - fitted(m1)))
```

3 Probe the Conditional Comparison

3.1 A conditional slope needs the covariance

At a centered baseline value w , the fitted practice slope is

$$\hat{\beta}_X(w) = \hat{\beta}_X + w\hat{\beta}_{XW}.$$

Because the two estimated coefficients are correlated, its variance is not obtained by adding their variances alone:

$$\begin{aligned} \text{Var}\{\hat{\beta}_X(w)\} &= \text{Var}(\hat{\beta}_X) + w^2 \text{Var}(\hat{\beta}_{XW}) \\ &\quad + 2w \text{Cov}(\hat{\beta}_X, \hat{\beta}_{XW}). \end{aligned} \tag{5}$$

The conditional-slope table applies this covariance formula to three meaningful baseline scores. Each interval uses $t_{.975,316} = 1.9675$.

Table 3

Conditional practice slopes at three supported baseline scores, with their declared targets.

Baseline score	Target slope	Fitted slope	SE	95% lower	95% upper
60	1.3	1.5484	0.1780	1.1982	1.8986
70	2.5	2.6972	0.1244	2.4524	2.9421
80	3.7	3.8461	0.1947	3.4630	4.2291

The estimated practice association rises from 1.5484 points per hour at baseline 60 to 3.8461 points per hour at baseline 80. These are three evaluations of one fitted surface, not three separate regressions and not evidence that students belong to three latent types.

The `emtrends()` function in the `emmeans` package calculates these conditional trends and their covariance-based intervals directly.

Agreement with the manual table is a computational check. The software does not choose defensible probe values. That responsibility remains with the analyst.

For the slope at centered baseline $w = 10$, applying the variance formula to the full-precision covariance matrix gives variance 0.0379 and SE 0.1947. Keep the entries at their stored precision until the final result is reported: early rounding of small covariance terms can alter the reconstructed SE.

Omitting covariance gives 0.1865. Adding the focal SE to ten times the product SE gives 0.2634. Neither computes the uncertainty of the fitted linear combination. A short contrast-vector calculation includes every needed covariance automatically:

```
b <- coef(m1); V <- vcov(m1)
cvec <- c(0, 1, 0, 10) # intercept, hours, baseline_c, product
sum(cvec * b)
sqrt(drop(cvec %*% V %*% cvec))
```

```
emmeans::emtrends(m1, ~ baseline_c, var = "hours",
  at = list(baseline_c = c(-10, 0, 10)))
```

A **simple slope** fixes the moderator and reads the focal comparison from the whole fitted model. For a two-hour comparison, multiply the slope and both endpoints by two: the fitted differences are 3.0969 at baseline 60 and 7.6923 at baseline 80. A different practice interval changes that multiplier, not the fitted relationship.

Now ask the interaction question directly: how much greater is the practice slope at 80 than at 60? Because both slopes use the same model, the focal coefficient cancels:

$$\hat{\beta}_X(10) - \hat{\beta}_X(-10) = 20\hat{\beta}_{XW} = 2.2977, \quad SE = 20 SE(\hat{\beta}_{XW}) = 0.2779.$$

The 95% interval is [1.7509, 2.8445]. This compares **two slopes**, whereas each row of Table 3 compares **one slope with zero**. Their uncertainty calculations answer different questions. In general, $\text{Var}(\hat{s}_2 - \hat{s}_1) = \text{Var}(\hat{s}_2) + \text{Var}(\hat{s}_1) - 2\text{Cov}(\hat{s}_2, \hat{s}_1)$. The two slope estimates share coefficients; subtracting interval endpoints or adding their SEs ignores that relationship.

```
20 * coef(m1) ["hours:baseline_c"]
20 * confint(m1) ["hours:baseline_c", ]
```

A large simple slope can coexist with zero interaction: both preparation levels might have the same strong practice association. Conversely, a changing slope can pass through zero at one baseline. Neither a test against zero at a chosen probe nor two separate significance labels replace a direct slope contrast (Gelman & Stern, 2006).

3.2 Choosing probe values and showing where the data are

The **pick-a-point** approach — also called the analysis of simple slopes — evaluates conditional slopes at a small set of substantively chosen moderator values (Aiken & West, 1991; Hayes, 2022). There is no rule that these values must be the mean and one standard deviation above and below it. Those values may be uninterpretable, outside the data, or different across studies. Better choices come from the measurement scale and the research question: policy thresholds, clinically or educationally meaningful cutoffs, prespecified percentiles, or values needed for an actual decision.

The 60, 70, and 80 probes work here because baseline score has an interpretable unit and all three values are well supported. A complete report also displays the moderator distribution rather than presenting the probes as if observations were equally dense everywhere.

3.3 The conditional-slope curve is the primary display

Three probe values are readings of one continuous object: the fitted conditional slope $\hat{\beta}_X(w) = \hat{\beta}_X + w\hat{\beta}_{XW}$ with its pointwise band, drawn over the moderator values the data actually support. That curve, not a boundary number, is the primary display for a continuous moderator (Bauer & Curran, 2005; Hayes, 2022). The marked zero-crossing boundaries are explained in Section 3.4. Their position relative to the observed support matters as much as their numerical value in Figure 2.

Read left to right, the curve tells one continuous story rather than three separate group findings. At the lowest supported baseline scores the fitted slope is slightly negative — about -0.4191 score points per hour — and its band covers zero. The band leaves zero at the upper boundary, 51.284, and the slope then rises steadily to about 5.6843 score points per hour at the highest observed baseline score. The three pick-a-point probes are three readings of this same curve. The band widens toward both ends because a conditional slope evaluated far from the centering value borrows heavily from the product coefficient, whose own uncertainty is then multiplied by a large w .

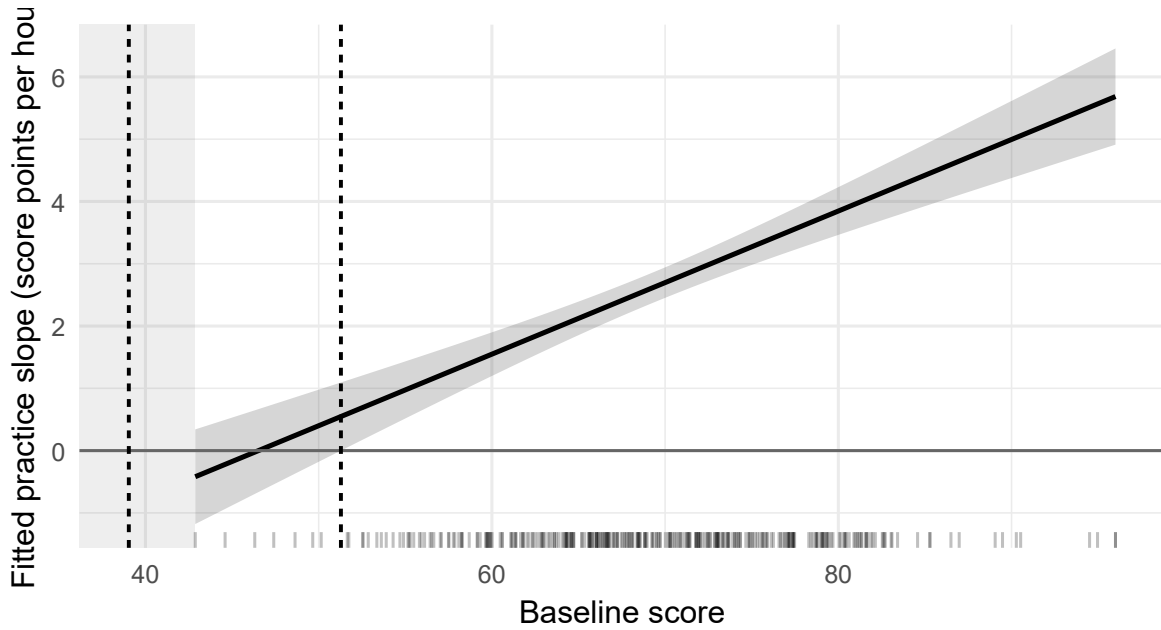


Figure 2: Fitted practice slope across the supported baseline-score range with a 95% pointwise band. Dashed vertical lines mark the two Johnson-Neyman boundaries, the shaded panel marks baseline scores below the smallest generated value, and the rug shows the observed baseline distribution.

Marking the boundaries on the curve also makes the support limitation visible rather than hidden in a table: the lower boundary sits inside the shaded panel, where no student was generated, so nothing in these data speaks to it.

3.4 Levels, cases and the limit of the algebra

A slope compares changes; a fitted mean describes a level. At four practice hours and baseline 70, the fitted mean is 70.4013 with mean-response interval [69.8644, 70.9382]. The new-case interval [60.8082, 79.9944] is wider because it also includes individual error variance. At baselines 60 and 80, the fitted means are 61.9836 and 78.8189. Precision about a mean does not imply precise prediction of each student.

A **Johnson-Neyman** calculation searches the moderator continuum for values at which a pointwise confidence interval for a conditional slope touches zero, and it labels the values on one side of those boundaries a **region of significance** (Bauer & Curran, 2005; Hayes, 2022; Johnson & Neyman, 1936). It can complement pick-a-point probes when the moderator is genuinely continuous and no small set of theory-based values captures the question. It replaces neither the conditional-slope curve, the magnitude estimates, nor the support audit.

For this fitted curve the pointwise interval excludes zero below baseline 39.039 and above 51.284; between these values it covers zero. Appendix A shows how the boundary equation and its sign determine those regions. The figure lets us check what those numbers mean before interpreting them.

This example reveals a support limitation that a pair of printed roots would conceal. The lower boundary, 39.039, lies below the smallest generated baseline score, 42.8747, so 0 of the 320 simulated students fall in the “significantly negative” region: it exists in the algebra and nowhere in the data. The upper boundary, 51.284, does lie inside the observed range, and 313 students (97.8125 percent of the sample) sit above it. The supported reading is therefore one-sided: across almost the entire baseline range these data cover, the practice slope is positive and separated from zero, and only a thin low-baseline sliver of 7 students falls in the interval-covers-zero region.

Any boundary is descriptive of that fitted model and its pointwise procedure. Scanning many moderator values raises multiplicity questions; a pointwise band is not a simultaneous band. A root outside the observed range has no empirical support, and a root inside a sparse tail may still be unstable. The phrase “region of significance” can also distract from magnitude and uncertainty. Pick-a-point answers questions at chosen values; Johnson-Neyman locates model-based zero-crossing boundaries. Neither method discovers a uniquely correct moderator cutoff, protects against specification search, or supplies simultaneous error control (Preacher et al., 2006).

A **simultaneous band** answers a wider question: does one band cover the entire conditional-slope curve? Under the correctly specified linear model with independent Normal errors of constant variance, a Scheffé band for the two coefficients determining that curve replaces 1.9675 by $\sqrt{2F_{.95;2,316}} = 2.4594$ in the estimate-plus-or-minus-SE calculation. It is wider because its coverage applies jointly to every linear combination of those two coefficients, hence every conditional slope. The two counts the dimensions of this slope family, not all four coefficients in the fitted mean model. Showing only the supported range makes the display useful; it does not validate extrapolation. Section 6.4 compares this band with pointwise intervals from two calculations.

4 Continuous by Categorical: Rank-Specific Slopes

4.1 The question, the records and the analysis sample

The sustained empirical question is:

Among the faculty of this college in 2008-09, how does the linear association between years of service and mean nine-month salary differ across the assistant, associate, and full professor ranks?

Each row is one faculty member. The outcome is nine-month salary, rescaled to thousands of dollars. The focal continuous predictor is years of service at the institution. Rank is a three-level categorical predictor. The target is a rank-conditional descriptive association and conditional mean prediction within observed support. It is not a return to loyalty, a promotion effect, or a claim about faculty salaries in general.

Data and software source

The `Salaries` data are called directly from the `carData` package (package DOI), which documents them as the 2008-09 nine-month salaries of the assistant, associate, and full professors at one U.S. college, collected as part of the administration's ongoing monitoring of salary differences between male and female faculty (Fox et al., 2022; Fox & Weisberg, 2019). This chapter uses only the package-distributed data and does not copy a local data file. One college in one year is not a probability sample of academia, so all inferential language below is bounded to these faculty under the working model.

Conditional trends are computed with the official `emmeans` package, whose `emtrends()` function evaluates fitted slopes from a model reference grid. The chapter also verifies key calculations from the fitted coefficient vector and covariance matrix.

The package dataset has 397 rows. The analysis rule is fixed before model fitting: retain a row only when salary, years of service, and rank are all observed, and use exactly the same rows for the additive and interaction models. Completeness is verified rather than assumed.

The fixed analytic sample contains 397 faculty members: 67 assistant professors, 64 associate professors, and 266 full professors. Here the complete-case rule excludes nothing, but stating and checking it is still part of a prespecified analysis: the audit is what shows the rule to be harmless rather than an untested assumption.

```
d0 <- carData::Salaries
dat <- d0[complete.cases(d0[c("salary", "yrs.service", "rank")]), ]
dat$rank <- droplevels(dat$rank)
dat$salary_k <- dat$salary/1000
dat$service_c <- dat$yrs.service - 5
```

4.2 Support first: where each rank actually lives

An interaction graph can tempt analysts to compare fitted lines where a rank has few or no observations. Support must therefore be examined within ranks, not only in the pooled sample.

Table 4

Years-of-service support and salary summaries by rank (thousand dollars).

Rank	n	Mini- mum	First quartile	Median	Third quartile	Maxi- mum	Mean salary	SD
Assistant	67	0	1	3	3	6	80.7760	8.1741
Associate	64	1	7	8	11	53	93.8764	13.8317
Full	266	0	15	21	30	60	126.7721	27.7187

The support asymmetry here is extreme, and it is the most important fact in this analysis. Assistant professors span only 0 to 6 years of service, while associate and full professors extend past fifty years. The range common to all three ranks is only 1 to 6 years. Whatever the model later reports about the assistant-professor slope is estimated from that six-year window; the model can say nothing about assistant professors with long service because none appear in these data. The dataset cannot establish whether such faculty existed at the college outside the recorded observations.

```
tapply(dat$yrs.service, dat$rank, range)
```

4.3 The centered model and its two product columns

The model centers years of service at 5:

$$S_c = S - 5.$$

This anchor lies in the observed range of all three ranks — it is inside even the narrow assistant-professor window. It is chosen so that rank lower-order coefficients represent salary differences at a supported common service value. It is not chosen to force a desired p-value or to make the ranks comparable in every other respect.

With assistant professor as the reference level, the model is

$$\begin{aligned} E(Y \mid S_c, R) = & \beta_0 + \beta_S S_c + \beta_A I(R = \text{AssocProf}) \\ & + \beta_P I(R = \text{Prof}) + \beta_{SA} S_c I(R = \text{AssocProf}) \\ & + \beta_{SP} S_c I(R = \text{Prof}). \end{aligned} \tag{6}$$

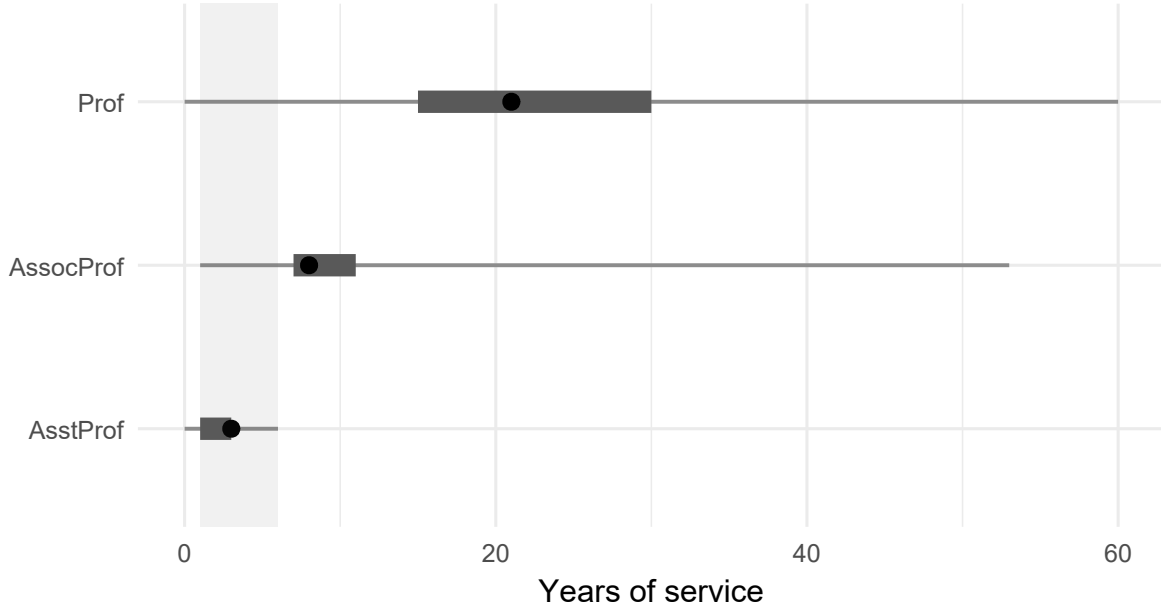


Figure 3: Rank-specific years-of-service support. Thick segments show interquartile ranges, thin segments show observed ranges, and points mark medians.

The three lines have the following interpretations:

$$\begin{aligned}
 \text{Assistant: } & \beta_0 + \beta_S S_c, \\
 \text{Associate: } & (\beta_0 + \beta_A) + (\beta_S + \beta_{SA}) S_c, \\
 \text{Professor: } & (\beta_0 + \beta_P) + (\beta_S + \beta_{SP}) S_c.
 \end{aligned}$$

The two product coefficients are slope differences from the assistant rank. They are not the associate and full-professor slopes themselves.

Table 5

Coefficients of the service-by-rank interaction model, read at their own anchors.

Conditional term	Estimate	SE	95% lower	95% upper
Assistant mean salary at 5 years	82.8231	5.8693	71.2837	94.3625
Assistant slope	0.7793	1.9452	-3.0450	4.6036
Associate — assistant at 5 years	13.7977	6.8835	0.2644	27.3310
Full — assistant at 5 years	46.0748	6.4445	33.4046	58.7449
Associate — assistant slope	-1.1740	1.9674	-5.0420	2.6940
Full — assistant slope	-0.8986	1.9492	-4.7309	2.9336

At five years of service, the fitted mean salary for assistant professors is 82.8231 thousand dollars. The fitted associate-minus-assistant difference at that service value is 13.7977 thousand dollars, with an interval from 0.2644 to 27.331. The fitted professor-minus-assistant difference is 46.0748 thousand dollars, with an interval from 33.4046 to 58.7449.

Those are conditional comparisons at one service value. They are not overall rank differences and are not comparisons at each rank's typical service. The product terms further show that the associate slope differs from the assistant slope by an estimated -1.174 thousand dollars per year, while the professor slope differs by -0.8986 — both with wide intervals, dominated by the six-year assistant support window.

```
s1 <- lm(salary_k ~ service_c * rank, data = dat)
coef(s1)
```

4.4 The two-degree-of-freedom product block

Because rank has three levels, its interaction with one continuous predictor contributes two columns. The appropriate nested comparison tests them jointly, and this omnibus block question comes before any examination of individual slopes or their pairwise differences.

Table 6

Classical and approximate robust tests of the two-column service-by-rank product block.

Variance procedure	Numerator df	Denominator df	F	p
Classical nested-model F	2	391	0.4860	.6155
Approximate HC3 joint Wald F	2	391	1.3138	.2700

The block comparison gives $F(2, 391) = 0.486$, $p = .6155$. Under the linear-model working assumptions, the data provide no clear evidence against one common service slope for the three ranks. That is a statement about what these data can distinguish, not a demonstration that the three slopes are equal — the later interval estimates show how wide a range of slope differences remains compatible. The test also does not say whether the lines are exactly linear or whether rank causally changes the salary return to service.

An approximate joint Wald sensitivity check using the HC3 covariance matrix — the estimator named in the chapter box, which keeps the ordinary least squares (OLS) coefficients that minimize the residual sum of squares but rebuilds their covariance from leverage-adjusted squared residuals instead of one pooled variance (Long & Ervin, 2000) — gives $F(2, 391) = 1.3138$, $p = .27$. Dividing the robust Wald statistic by the number of restrictions and comparing it with an F distribution using the residual degrees of freedom is a transparent finite-sample approximation, not an exact robust F reference result. The robust and classical block procedures

therefore support the same bounded conclusion here, although the p-values differ because the procedures use different variance models, not because the fitted slopes changed.

The order of `anova(reduced, full)` matters for readable output and for the narrative. Both models must use the same response and exactly the same rows. If the full model silently loses more observations because a new variable is missing, the models are not nested on a common dataset and the comparison no longer answers the intended block question. Here the analysis sample was fixed once, before fitting, so the printed row counts agree by construction.

```
s0 <- lm(salary_k ~ service_c + rank, data = dat)
anova(s0, s1)
```

4.5 Rank-specific slopes and their comparisons

With the joint block assessed, the rank slopes are more direct than the reference-coded product coefficients. `emtrends()` obtains the fitted service trend within each rank.

Table 7

Rank-specific fitted service slopes with covariance-based standard errors.

Rank	Slope (thousand dollars per year)	SE	95% lower	95% upper
Assistant	0.7793	1.9452	−3.0450	4.6036
Associate	−0.3947	0.2949	−0.9744	0.1850
Full	−0.1193	0.1253	−0.3656	0.1270

Within the observed rank-specific ranges, one additional year of service is associated with an estimated mean salary difference of about 0.7793 thousand dollars for assistant professors, −0.3947 for associate professors, and −0.1193 for full professors. Every classical interval includes zero, and the assistant-professor interval is several times wider than the others — its slope is estimated from a six-year window. The rank, unit, support and associational reading belong together. The slightly negative continuing-rank estimates are compatible with cohort and market effects that these cross-sectional data cannot separate from service itself.

Pairwise slope differences address a different family of questions. We use a Tukey-style adjustment for the three pairwise comparisons; the prespecified omnibus block test reported above remains the primary check of whether slopes vary somewhere across ranks.

The adjusted p -values are 0.8220, 0.8895 and 0.6661 for associate-minus-assistant, full-minus-assistant and full-minus-associate slopes. The multiplier is 2.3527 on 391 degrees of freedom. This calibration is approximate for unequal slope-estimation variances, rather than the exact balanced one-way Tukey result. No pairwise slope contrast is distinguishable from zero, in agreement with the omnibus block. This coherent presentation — an omnibus check, then interval estimates for every slope and contrast — is more informative than announcing that one reference-coded product term is small and another is not.

```
emmeans::emtrends(s1, ~ rank, var = "service_c")
```

4.6 Coordinates, common-value contrasts and the picture

To verify that claim, refit the salary model with full professor as the reference rank. Coefficient labels and lower-order interpretations change, but the fitted surface does not.

The maximum fitted-value difference is numerical noise. Reference coding changes which line is described directly by β_0 and β_S and which slope differences appear as product coefficients. It does not change the three lines, residuals, predictions, or the interaction-block test.

The same principle applies to centering years of service. Had we centered at 3 instead of 5 years, rank lower-order coefficients would compare fitted means at 3 years. The slopes and slope differences would remain the same. An analyst should choose the anchor before looking for a more favorable coefficient table.

Table 8

Four conditional questions and one block question, each with its own quantity.

Question	Quantity	Salary example
How does salary vary with service within a rank?	Conditional slope	−0.1193 thousand dollars per year for full professors
How do ranks differ at one common service value?	Conditional mean contrast	Full − assistant 46.0748 at 5 years
What mean salary does the model fit for a rank and service value?	Conditional mean	Associate 96.6208 at 5 years
What range is plausible for one new case under the model?	New-case interval	[49.6127, 143.6288] for one associate at 5 years
Is one common slope adequate across the ranks?	Product-block test	$F(2, 391) = 0.486$

At two, three, five and six service years, the fitted full-minus-assistant salary contrasts are 48.7706, 47.8720, 46.0748 and 45.1762 thousand dollars, with SEs 4.2152, 4.2532, 6.4445 and

8.0405. These are linked readings of one fitted surface, all inside the common service window. Their uncertainty grows toward the edge of the short assistant range.

A single grid from the pooled minimum to pooled maximum would draw the assistant-professor line across fifty unsupported years. The plotted prediction grid uses the minimum and maximum service within each rank.

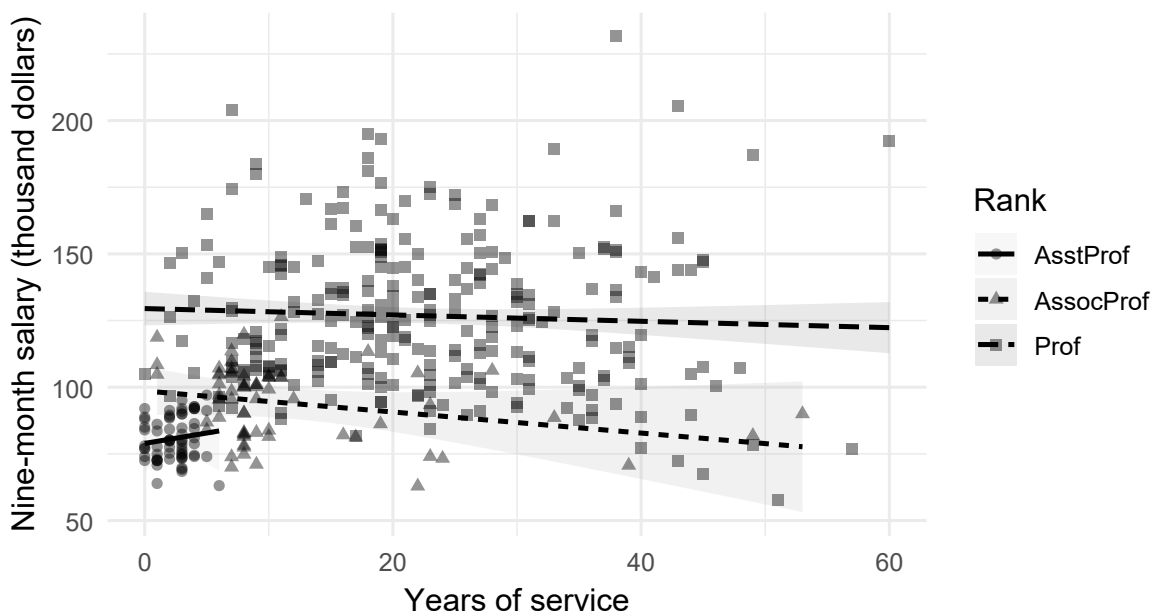


Figure 4: Rank-specific fitted mean salary with 95% mean-response bands. Each line and band stop at that rank's observed service limits.

The visual and the trend table agree: rank differences in salary level dominate, the fitted within-rank service trends are close to flat, and the assistant-professor line is both short and imprecisely estimated. The graph also keeps a limitation visible that a coefficient table can hide: the ranks occupy drastically different parts of the service range. The model estimates each within-rank slope from its own support; a cross-rank comparison at a particular service value is most credible inside the narrow common support of 1 to 6 years.

```
predict(s1, newdata = data.frame(service_c = 0,
  rank = factor("AssocProf", levels = levels(dat$rank))),
  interval = "confidence")
```

5 Factor by Factor: Cell Means Revisited

5.1 Two factors: the same nonadditive question

When both predictors are factors, an interaction is nonadditivity among cell means. The `carData::Moore` data provide a compact illustration from social psychology: 45 participants in a conformity experiment (Moore & Krupat, 1971), with the number of conforming responses as the outcome, the manipulated status of a partner (low or high) as one factor, and the participant's measured authoritarianism category (low, medium, or high F-scale score) as a three-level factor. The package documentation describes partner status as manipulated but does not document random assignment or the full experimental protocol; the example therefore treats it as an experimental-condition contrast rather than assuming causal identification from the dataset description alone. This example is separate from the salary analysis and is not needed for its conclusions. Under the classical common-variance linear model, its product block is statistically detectable; the conclusion is sensitive to the variance model and outcome scale, as shown below. The cell summary reports, for each of the six cells, its size, its mean number of conforming responses, and its standard deviation (SD), the typical distance of a participant's score from that cell's mean.

Table 9

Conformity cell counts, means and standard deviations.

Authoritarianism	Partner status	<i>n</i>	Mean conformity	SD
Low	Low	10	8.9000	2.6437
Low	High	5	17.4000	4.5056
Medium	Low	4	7.2500	3.9476
Medium	High	11	14.2727	3.9520
High	Low	8	12.6250	7.3473
High	High	7	11.8571	3.9340

An additive factor model requires the partner-status difference to be the same at every authoritarianism level. The product model allows that difference to vary. Its factor levels are ordered low, medium, high for authoritarianism and low, high for partner status.

5.2 The block, simple differences and interaction contrast

The block has two degrees of freedom because a three-level authoritarianism factor contributes two indicator columns and therefore two product columns. No middle category is discarded to manufacture a 2-by-2 table. The classical test gives $F(2, 39) = 4.1846$, $p = .0226$, whereas the approximate HC3 joint Wald test gives $F(2, 39) = 2.8294$, $p = .0712$. The cells are small and unbalanced, their spreads differ, and conformity is a bounded count. Ordinary linear model

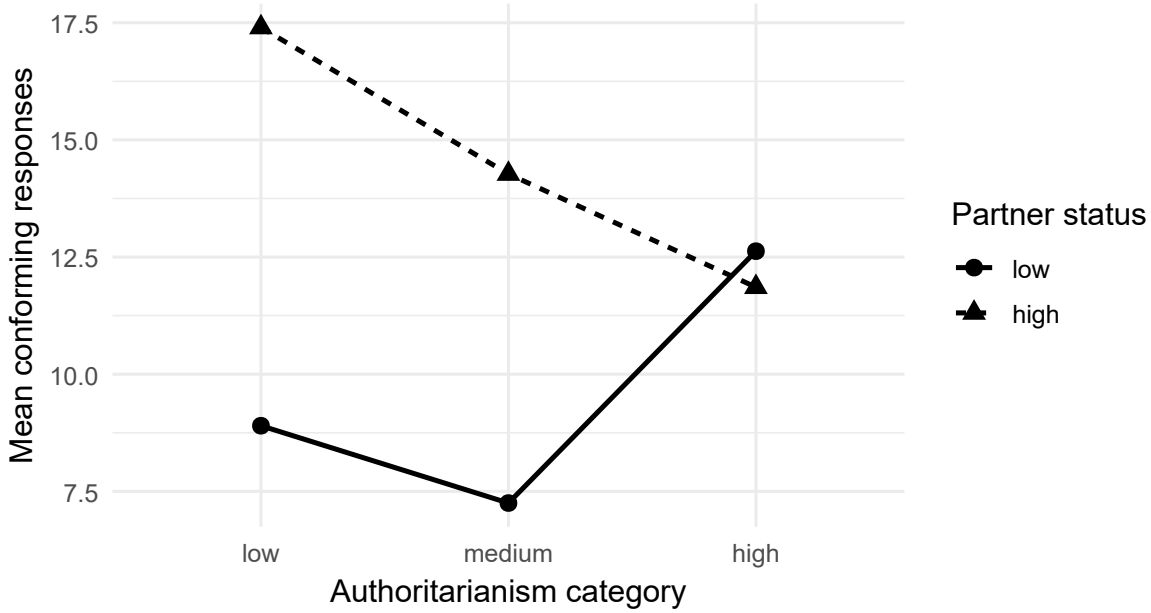


Figure 5: Moore conformity cell means by partner status and authoritarianism category. The status difference is large at low and medium authoritarianism and essentially absent at high authoritarianism, a crossing nonadditive profile.

inference is therefore a working approximation here, not a scale-robust verdict. The cell-mean pattern remains pedagogically useful; the strength of inferential evidence must be reported as variance-model-sensitive.

For any two authoritarianism categories, a **statistical difference-in-differences (DiD)** — the difference between two group differences — is an interaction contrast. Comparing the low and high categories gives

$$\begin{aligned} \hat{C}_{\text{DD}} = & (\bar{y}_{\text{low},H} - \bar{y}_{\text{low},L}) \\ & - (\bar{y}_{\text{high},H} - \bar{y}_{\text{high},L}), \end{aligned}$$

where H and L denote high- and low-status partners.

The contrast is 9.2679 conforming responses: the partner-status difference is about 8.5 responses among low-authoritarianism participants and about -0.7679 among high-authoritarianism participants. Algebraically it is a product-term contrast. Partner status is described as manipulated, but the package documentation alone does not establish random assignment, implementation fidelity, or the absence of post-assignment complications. The status comparison should therefore be described as an experimental-condition contrast unless the original study documentation supplies those warrants. Authoritarianism is measured rather than assigned, so

any variation of the status contrast across F-scale categories also depends on observational group comparability. In a policy or before-after study, a causal difference-in-differences interpretation would additionally require a defensible treatment and timing design, a counterfactual trend argument such as parallel trends, absence of relevant anticipatory or spillover behavior, and uncertainty appropriate to the assignment and dependence structure. Those design-based methods are outside this chapter.

5.3 What the difference of differences does not carry

Statistical interaction is not a causal design

A product term describes nonadditivity in a model. Causal interpretation comes from design and assumptions outside the coefficient algebra. The same product term can appear in an observational cross-section, a randomized factorial experiment, or a longitudinal policy comparison; those settings do not inherit the same causal meaning.

```
moore <- carData::Moore
with(moore, tapply(conformity, list(fcategory, partner.status), mean))
c1 <- lm(conformity ~ partner.status * fcategory, data = moore)
```

For low versus high authoritarianism, the interaction contrast is

$$\widehat{C}_{DD} = (\bar{y}_{\text{low},H} - \bar{y}_{\text{low},L}) - (\bar{y}_{\text{high},H} - \bar{y}_{\text{high},L}) = \mathbf{c}^T \widehat{\boldsymbol{\beta}} = 9.2679, \quad SE = \sqrt{\mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}} = 3.4507.$$

The classical pointwise interval is [2.2883, 16.2475]. If this comparison was selected after inspecting the cell pattern, the comparison family needs a simultaneous calibration as in Chapter 4; an attractive contrast is not retroactively prespecified.

6 Check, Qualify and Resample

6.1 Where the model is likely to strain

The salary interaction model assumes an adequately linear conditional mean within each rank, independent information from the modeled rows, and sufficient joint service-by-rank support. Classical standard errors and prediction intervals additionally use a common residual variance; small-sample exact t and F results use a conditional Normal-error model. Leverage and influence matter because product columns can make sparse predictor combinations especially consequential. Residual plots do not certify these assumptions; they show where parts of the approximation may be incomplete (Fox, 2016; Gelman et al., 2020).

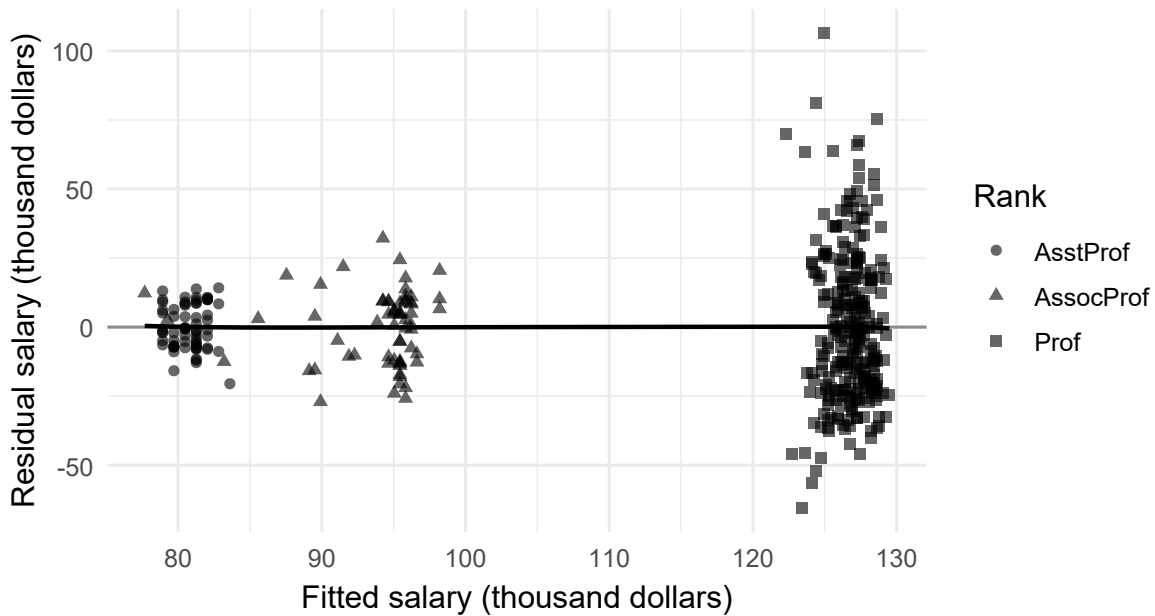


Figure 6: Residuals versus fitted values for the salary interaction model. Points are marked by rank and the smooth summarizes remaining mean pattern.

The residual-versus-fitted plot should be read with the raw plot and support display. Ranks occupy different fitted ranges, and residual spread is visibly larger at professor-level fitted values, so the common-variance working assumption deserves the HC3 sensitivity check below rather than silent acceptance.

The Q-Q plot is an approximation check, not a requirement that the salaries themselves be Normally distributed. The linear-model assumption concerns the conditional errors used for exact small-sample reference results.

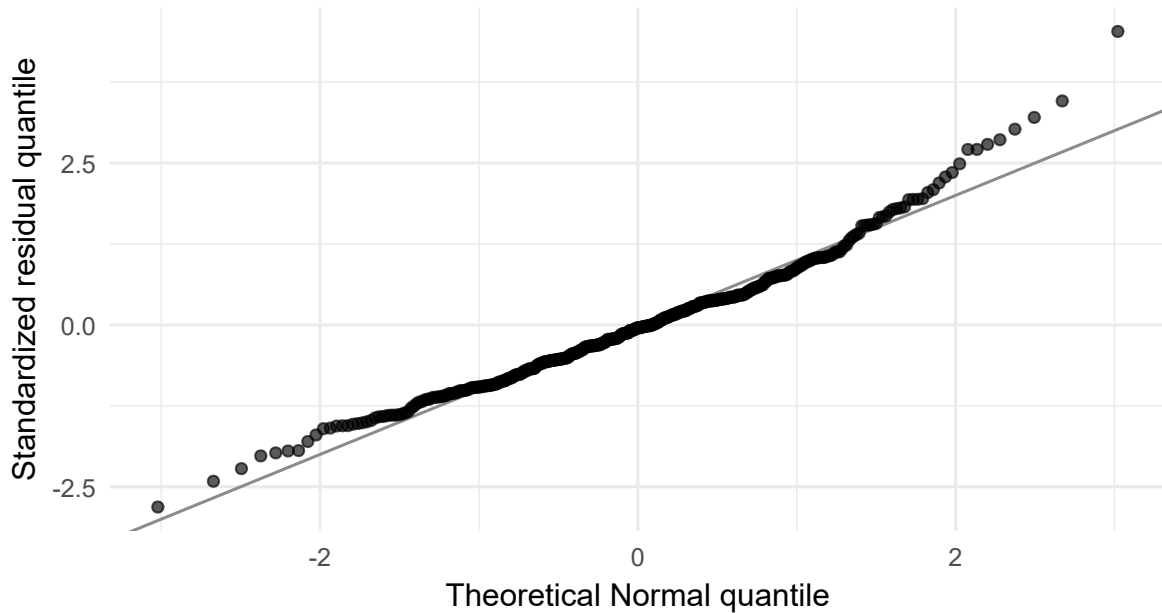


Figure 7: Normal quantile plot of standardized residuals from the salary interaction model. Tail departures matter most for small-sample interval calibration and unusual cases.

```
tapply(residuals(s1), dat$rank, sd)
plot(fitted(s1), residuals(s1))
abline(h = 0, lty = 2)
```

6.2 The same slopes under a different covariance

The HC3 standard errors introduced with the block test replace the common-variance coefficient covariance matrix with a leverage-adjusted, residual-based estimate while retaining the OLS coefficients and the fitted mean model (Long & Ervin, 2000; Zeileis, 2004). Standard errors, approximate intervals, and approximate joint Wald tests derived from that matrix can change. Applied to the rank-specific trends, they are a sensitivity analysis for uncertainty, not a repair for a wrong conditional mean and not a license to ignore dependence.

Table 10*Rank-specific slopes under the classical and HC3 coefficient covariances.*

Rank	Slope	Classical SE	HC3 SE	Approximate HC3 lower	Approximate HC3 upper
Assistant	0.7793	1.9452	0.8686	−0.9284	2.4870
Associate	−0.3947	0.2949	0.1679	−0.7248	−0.0645
Full	−0.1193	0.1253	0.1884	−0.4898	0.2512

Here the two variance treatments disagree in an instructive way. The HC3 standard errors are smaller for the two junior ranks — whose residual spread is below the pooled level — and larger for full professors. The approximate HC3 sensitivity interval for the associate-professor slope excludes zero while the classical interval does not. A borderline conclusion that moves when the variance model changes should be reported as sensitive, not resolved by quietly choosing whichever calculation looks better. There is a second trap here. Under HC3 the associate slope interval excludes zero and the full-professor interval covers zero. That does not show that the two slopes differ. Their directly estimated associate-minus-full contrast is -0.2754 , with approximate HC3 SE 0.2524 and interval $[-0.7716, 0.2209]$ using the same residual-df reference ($p = .2759$). The interval leaves substantial uncertainty about the direction of the slope difference. Comparing two labels discards the very covariance that the contrast needs; its nonrejection also does not establish equal slopes.

Neither calculation repairs the fitted mean, the design, or the dependence structure.

```
V <- sandwich::vcovHC(s1, type = "HC3")
cvec <- c(0, 1, 0, 0, 1, 0) # associate slope
sqrt(drop(cvec %*% V %*% cvec))
```

6.3 Influence and a competing explanation

The most influential row is a full professor with 60 years of service and a salary of 192.253 thousand dollars — a long-service, high-leverage point at the far edge of the professor range. Omitting it moves the professor slope slightly and leaves the other slopes essentially unchanged; the largest Cook’s distance is small, and the broad pattern of imprecise, overlapping slopes remains. This leave-one-out fit is a sensitivity description. Deletion would require a documented recording error, an explicit target-population rule, or another substantive reason independent of Cook’s distance.

If the true relationship is curved and ranks occupy different ranges, separate straight lines can appear to have different slopes even when the same curved function applies. A compact sensitivity model adds one common quadratic service term while retaining the full lower-order interaction structure.

This one-degree-of-freedom check provides little evidence that a shared quadratic term materially improves the fitted mean. It does not prove that every rank relationship is exactly linear. Rank-specific curvature would require additional terms and, for assistant professors, substantially more support than a six-year window can offer. A sensitivity model should clarify a plausible failure mode, not become an unrestricted search for significance.

```
which.max(cooks.distance(s1))
anova(s1, lm(salary_k ~ service_c * rank + I(service_c^2), dat))
```

6.4 Bootstrap the conditional comparison

Resampling follows the question. For the generated students, that question is the practice slope at baseline 60, 70 or 80, or the difference between two such slopes. One complete-row resample supplies one new fit, and that fit supplies **all** these answers. Resampling each data column separately would destroy the observed association. Resampling the two slopes separately would destroy their shared estimation uncertainty.

Treating the generated student rows as independent and identically distributed, draw 320 complete rows with replacement and refit the same hierarchical model. Repeat 2,000 times with seed 90910. This approximates repeating the sampling and fitting process for the stated slope estimands; it does not generate new independent evidence (Efron & Tibshirani, 1993; Freedman, 1981). The three distributions in Figure 8 contain estimates, not individual students' scores. Their centers and widths differ because the three probes ask different conditional questions of the same records.

Table 11

The same three simple slopes followed through 2,000 complete-row resamples.

Baseline	Original slope	Bootstrap SE	Percentile lower	Percentile upper
60	1.5484	0.1764	1.1866	1.8790
70	2.6972	0.1262	2.4532	2.9328
80	3.8461	0.1954	3.4523	4.2187

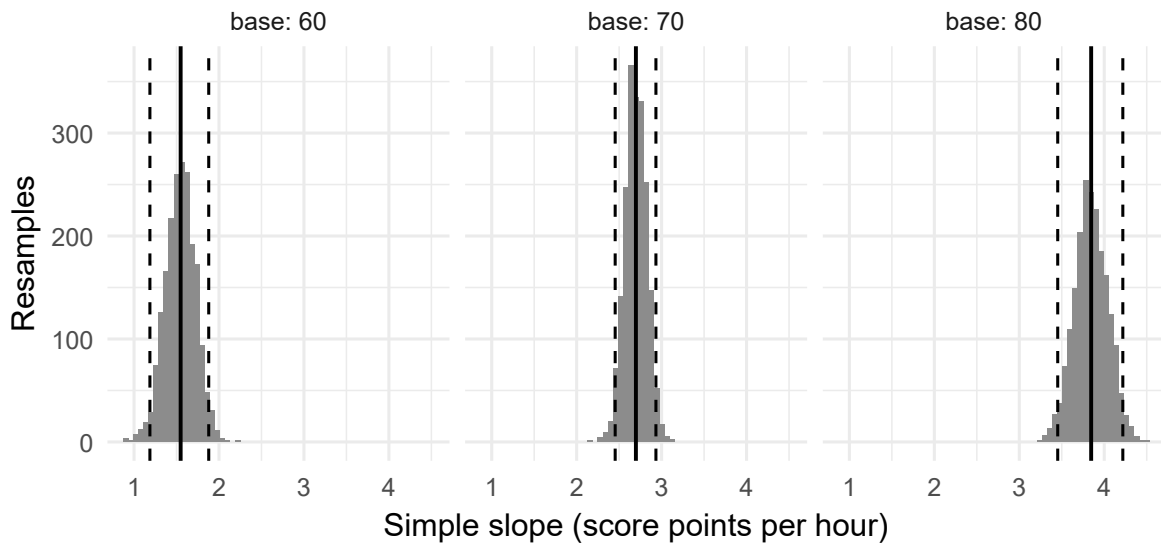


Figure 8: Bootstrap distributions of the three conditional practice slopes. Solid lines mark original estimates; dashed lines mark the 2.5th and 97.5th percentiles. Each panel uses the same 2,000 row resamples, with a shared horizontal scale for comparing their spreads.

For every resample, subtract its slope at 60 from **that same resample's** slope at 80. The resulting difference has bootstrap SE 0.2737 and percentile interval [1.7404, 2.8368], close to the classical interval [1.7509, 2.8445]. If `draws` holds one slope per row and one resample per column, the calculation is short:

```
diff <- draws[3, ] - draws[1, ]
sd(diff)
quantile(diff, c(.025, .975))
```

The agreement here is expected from a deliberately simple, well-behaved generating model. It is not a general promise that bootstrap intervals are wider, more conservative or more accurate. A percentile interval reads quantiles of a resampling distribution; it is not a statement that this realized interval contains the true slope with 95% posterior probability. More resamples make those quantiles less sensitive to Monte Carlo variation. They do not make the study's 320 observations more informative.

Figure 9 answers another question: what changes when the coverage statement concerns a whole curve? All panels use the same original estimate. The first two show **pointwise** intervals: one classical, one percentile bootstrap. The third uses Section 3.4's classical simultaneous multiplier and is visibly wider. Its additional protection concerns the family of slopes, under the

classical model assumptions; it is not a correction for a wrong model or weak design. Taking percentiles at many grid points does not turn the second panel into a simultaneous band.

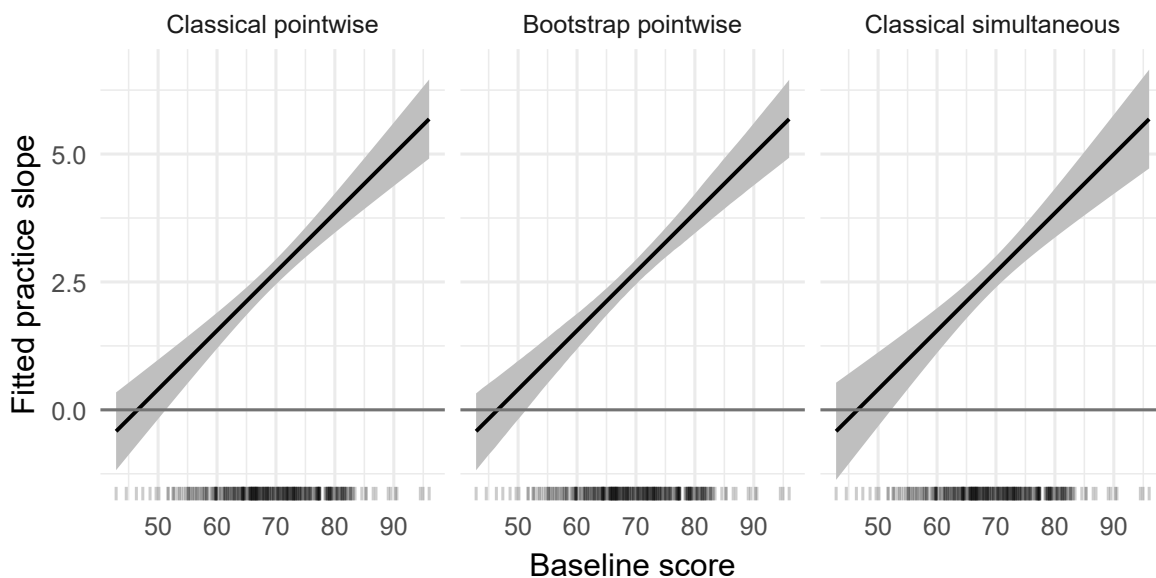


Figure 9: One slope curve, three coverage statements. Classical and bootstrap pointwise bands concern a fixed moderator value; the wider classical Scheffé band simultaneously covers the slope family under its stated linear-model assumptions. All displays stop at observed support.

Return now to the salary example, where unequal spread made the choice of uncertainty calculation consequential.

The associate slope is the quantity whose reading changed under HC3. A case bootstrap asks what another sampling approximation says about it: draw whole faculty records with replacement and refit the same model. This preserves the empirical association of rank, service and spread, while treating rows as independent and exchangeable. With seed 90909, all 2,000 resamples here contain all three ranks. The bootstrap machinery is the case-resampling scheme of Chapter 9 (Efron & Tibshirani, 1993; Freedman, 1981).

Table 12

Three uncertainty calculations for the same three conditional slopes (seed 90909, $B = 2000$).

Rank slope	Estimate	Classical SE	HC3 SE	Bootstrap SE	Percentile 2.5%	Percentile 97.5%
Assistant	0.7793	1.9452	0.8686	0.8074	−0.7172	2.3841
Associate	−0.3947	0.2949	0.1679	0.1736	−0.8066	−0.1436
Full	−0.1193	0.1253	0.1884	0.1872	−0.4928	0.2522

The associate bootstrap SE 0.1736 is close to HC3’s 0.1679 rather than the classical 0.2949. Its percentile interval also excludes zero. The resampling skewness is -0.8463 , so a symmetric normal approximation, $[-0.7350, -0.0544]$, conveys a somewhat different shape than the percentile interval $[-0.8066, -0.1436]$. The interval method belongs in the report. Neither interval is an independent replication of the study.

If `draws` holds one bootstrap associate-slope estimate per element, these two operations summarize that resampling distribution:

```
sd(draws)
quantile(draws, c(.025, .975))
```

Row resampling cannot represent unknown departmental dependence, create long-serving assistant professors, repair measurement problems, or identify a promotion effect. A design with clusters, strata or weights requires a resampling scheme that follows them. Agreement between HC3 and the case bootstrap is a sensitivity result about these same observations.

7 From Analysis to a Defensible Claim

7.1 The pattern before the verdict

The interaction block answers whether one common slope is adequate under the fitted linear model. A defensible report does not stop there. It describes the conditional pattern, gives interval estimates in original units, shows where the comparisons are supported, and states what the design does not identify.

For the salary analysis, the main pieces are:

- 397 faculty members at one college in 2008-09, with rank-specific service ranges shown explicitly;
- a hierarchical model with salary in thousands of dollars, service centered at 5 years, and a two-degree-of-freedom product block;
- an omnibus interaction comparison of $F(2, 391) = 0.486$, $p = .6155$;

- estimated within-rank slopes with confidence intervals;
- fitted means or contrasts at a supported common service value;
- residual, unequal-variance, influence, curvature, and support checks;
- three uncertainty procedures for the same conditional slopes, including the one slope whose approximate sensitivity interval excludes zero under HC3 and case resampling but not under the classical common-variance model; and
- an observational interpretation bounded to the represented data.

A bounded report

Among the 397 faculty members of this college in 2008-09, rank differences in mean nine-month salary were large, but the data provided no clear evidence that the linear service-salary association differed across ranks ($F(2, 391) = 0.486, p = .6155$). Within each rank's observed range, the estimated slopes were 0.7793 thousand dollars per service year for assistant professors (95% confidence interval $[-3.045, 4.6036]$), -0.3947 for associate professors ($[-0.9744, 0.185]$), and -0.1193 for full professors ($[-0.3656, 0.127]$). Every classical interval includes zero and the intervals overlap; the assistant-professor slope rests on a service range of only 0 to 6 years. The associate-professor slope was the one quantity sensitive to the variance procedure: its approximate sensitivity interval excluded zero under both an HC3 variance model ($[-0.7248, -0.0645]$) and a 2,000-resample case bootstrap ($[-0.8066, -0.1436]$) but not under the classical common-variance model, and its resampling distribution was skewed, so the interval type matters as well. The result is a descriptive conditional association for these faculty; it does not estimate a return to service, identify promotion or cohort mechanisms, or support extrapolation beyond observed rank-service combinations.

This paragraph reports a null block result without converting it into a claim that the slopes are equal, and it reports the variance-model sensitivity instead of quietly resolving it. Reporting what the data cannot distinguish is as much a finding as reporting what they can.

7.2 A reusable template and the statements it blocks

A reusable interaction report can be organized in six sentences.

1. **Target and unit:** “Among [analytic units], we examined whether the [outcome-unit] association between [focal predictor] and [outcome] varied across [moderator values or groups].”
2. **Model and support:** “We fit a hierarchical [model scale] model on [fixed sample], with probes restricted to [observed joint support].”
3. **Block evidence:** “Adding the [named] product block changed model fit by [degrees of freedom, statistic, interval or p-value].”
4. **Conditional estimates:** “At [prespecified supported values], the estimated slopes or contrasts were [estimates and intervals in units].”

5. **Checks:** “Residual, variance, influence, curvature, and alternative- coding checks suggested [specific consequences].”
6. **Boundary:** “The result supports [bounded associational or predictive conclusion] but does not establish [causality, mechanism, generalization, or unsupported extrapolation].”

Not every paper must use these exact sentences. Every report should make their information recoverable.

Table 13

Common interaction overclaims and their defensible replacements.

Attractive statement	Why it fails	Defensible replacement
“The interaction proves moderation.”	A product term is model-based nonadditivity; substantive moderation needs a theory-defined role and a design.	“The fitted conditional slope varied across the named values or groups.”
“There is no effect of X .”	The lower-order coefficient is the slope only at the coded zero or reference value.	“At $W = 0$ under this coding the estimated slope of X was ...”
“The lines interact only if they cross.”	Interaction means nonparallelism on the modelled scale.	“The estimated slopes differed over the displayed support.”
“Centering fixed the collinearity and the confounding.”	Centering changes the origin, not the information, the omitted variables, or the design.	“Centering made the zero-value comparison interpretable.”
“The product p -value shows where the effect exists.”	One block test does not describe conditional magnitudes across the moderator.	“The slope curve with its band and support shows where the comparison is estimated and how precisely.”
“One group’s slope is significant and another’s is not, so they differ.”	Two significance verdicts are not a test of their difference.	“The slope contrast was ... with interval ...”
“A large product p -value shows the slopes are equal.”	Nonrejection is not evidence of no difference.	“A slope difference between ... and ... remains compatible with these records; claims are restricted to observed joint support.”

The sustained salary example is exactly this situation, so it deserves its own statement of principle. A large product p -value is not proof that conditional slopes are exactly equal. It says the data and model do not distinguish the product block from zero with the chosen precision and reference procedure. A bounded null report gives the estimated difference and its interval, then asks whether values of practical importance remain compatible with the data.

For example, suppose an estimated slope difference is 0.08 outcome units per predictor unit with a 95% interval from -0.12 to 0.28 . “No interaction” hides the fact that modest negative and positive differences remain compatible. The relevant next question is substantive: would any part of that interval change the decision or theory?

Why can a study estimate a large average association while leaving moderation imprecise? A product coefficient needs information about how comparisons vary across **joint predictor combinations**. Consider the same 320 generated students. Keep practice hours and each student’s original generated error fixed, compress the centered baseline scores to 30% of their original deviations, and generate scores again from the same equation, including the true product coefficient 0.12. This changes the design’s predictor coverage, not the true interaction or error SD.

Predict before reading the intervals: would retaining 320 students and the same error realization preserve the precision of the interaction estimate? Which part of the information has changed?

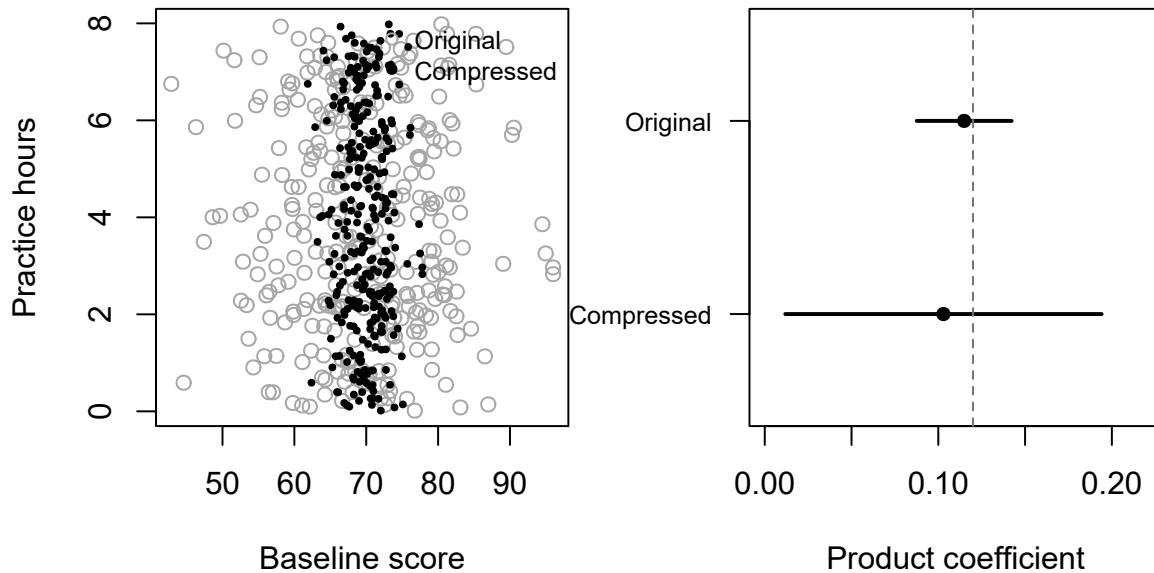


Figure 10: The same sample size and true interaction, with different predictor coverage. Left: compression narrows baseline variation while preserving all practice values. Right: the product-coefficient interval widens around the unchanged generating coefficient 0.12 (dashed line).

Baseline support contracts from 42.8747–96 to 61.8624–77.8000. The product estimate changes from 0.1149 (SE 0.0139, $[0.0875, 0.1422]$) to 0.103 (SE 0.0463, $[0.0118, 0.1941]$). Both samples arise from a true coefficient of 0.12, yet the compressed design leaves much more uncertainty about its size. The same row count is not the same information. In this controlled comparison

the SE is multiplied by $1/0.3$; that numerical relation uses the shared predictor values and errors, not a universal rule for every restricted dataset.

Sparse tails, imbalanced groups and noisy measurements can similarly weaken an interaction comparison (Gelman et al., 2020). A single simulated sample does not estimate statistical power. It shows what a design calculation should investigate: which effect sizes and intervals could this joint predictor distribution resolve? More bootstrap repetitions cannot replace missing predictor variation. Probing baseline 60 in the compressed sample would also be extrapolation, although it was supported in the original sample.

7.3 The research record for this analysis

Table 14

Closing research record for the faculty-salary interaction analysis.

Element	Closing answer for the faculty-salary analysis
Question	Whether the service-salary association varies across academic ranks, rather than assuming one common slope or searching over possible products
Records and design	One faculty member per row, one U.S. college, one academic year; rank and service were observed, not assigned, and no broader sampling scheme is documented
Measurement	Nine-month salary and years of service are recorded administrative variables; they do not measure productivity, market conditions, or the mechanisms behind a salary structure
Estimand and support	Rank-specific fitted service slopes and their differences over each rank's own support; cross-rank contrasts are credible only inside the 1-to-6 year common window
Model and estimation	One hierarchical OLS model with two product columns, service centered at a supported value, and every reported quantity built as a linear combination of the coefficients
Uncertainty and checks	The additive restriction compared by a two-degree-of-freedom block test; classical, HC3 and case-bootstrap uncertainty for the same slopes; residual form, influence, curvature, support and scale examined and reported
Bounded claim	These records do not clearly distinguish three rank-specific slopes from one common linear association; the weak interaction evidence is a bounded association, not a moderator mechanism

The record includes the package datasets, complete-case rule, scales, model formulas, anchors, probes and resampling unit. The generated example uses seed 909 with 320 rows. The practice

and salary bootstraps use seeds 90910 and 90909 respectively, each with 2,000 whole-row resamples. Reproducibility recovers the numerical calculation; the design record establishes which interpretation it can carry.

8 Reading Bridge

A product term releases a common-slope restriction. Lower-order coefficients are comparisons at stated anchors; centering and releveling move the coordinates without changing the fitted surface. Conditional slopes, group contrasts and differences of differences all use coefficient covariance for their uncertainty. The block test examines additivity; curves, intervals and support show what the comparison means.

The salary records illustrated an honest weak-interaction result and one slope sensitive to the variance assumption. Chapter 11 changes the outcome to binary and the mean scale to the logit. The conditional question continues, but interaction on the linear-predictor and probability scales will become different statements about the same fit.

Key terms

Table 15

Concepts introduced here.

Term and common alternatives	Meaning or essential distinction
Product term (interaction term; multiplicative term)	A column formed by multiplying two predictors, so that each one's fitted comparison depends on the other's value. Its coefficient is a difference in slopes, not a slope, and nonadditivity is defined on the modelled scale: a nonlinear change of outcome scale can create or remove it.
Conditional slope (simple slope; conditional effect)	The fitted difference in outcome per one-unit difference in a focal predictor evaluated at a stated value or group of the other predictor. It is a linear combination of coefficients, so its SE uses their covariance, and it describes a conditional mean, not an individual case.
Moderator (moderating variable; moderation)	The variable across whose values or groups a named focal relationship is allowed to vary. The product-term algebra is symmetric in the two predictors, so the label comes from the research question; a moderation reading describes when a relationship differs, not how an effect is transmitted.

Term and common alternatives	Meaning or essential distinction
Hierarchy principle (marginality; well-formulated model)	The rule that a model containing a product also contains both component terms. Dropping a component forces the other's slope to zero at its coded zero and makes the fitted surface depend on the origin; hierarchy does not require that every possible product be added.
Centering a predictor (anchoring; reparameterization)	Replacing a predictor by its difference from a chosen value, so that coefficients anchored at zero become comparisons at that value. It changes the intercept and the lower-order coefficients and leaves the product coefficient, the fitted values, the residuals and the product-block test unchanged; it repairs no design, selection, or measurement problem.
Probe value (pick-a-point value)	A stated value of the moderator at which a conditional slope or contrast is evaluated, chosen from the measurement scale, a threshold, or a decision need rather than by an automatic mean-and-SD rule, and restricted to values the records support.
Johnson–Neyman boundary (region of significance)	A moderator value at which a pointwise interval for a conditional slope touches zero, found as a root of a quadratic whose leading coefficient decides which side is the region. A boundary outside the observed support describes the algebra and not the data, and a pointwise band is not a simultaneous band.
Interaction contrast (difference in differences within a model)	A linear combination of coefficients comparing two conditional comparisons, such as one group's difference minus another group's difference. Under reference coding it equals a product coefficient up to sign; its SE uses the coefficient covariance, and its causal reading depends on design, not on the algebra.

Further reading

Aiken and West (Aiken & West, 1991) and Cohen and colleagues (Cohen et al., 2003) develop centering and conditional comparisons. Bauer and Curran (Bauer & Curran, 2005) and Preacher and colleagues (Preacher et al., 2006) explain probing and significance boundaries. Fox and Weisberg (Fox & Weisberg, 2019) supply the data and regression software route. The emmeans interaction vignette connects conditional comparisons to implementation.

Exercises

Exercise 1: Read the graph before the table (Foundational)

A study plots a continuous outcome against weekly practice hours for two program groups. Over the observed range of 0 to 6 hours, the fitted Group A line rises from 50 to 62 and the fitted Group B line rises from 58 to 76. The confidence bands overlap at some practice values and the lines do not cross. (1) Calculate each fitted slope in outcome units per hour and their difference. (2) Give the fitted group gap at 0 hours and at 6 hours, and the difference between those two gaps. (3) Describe the pattern as additive or nonadditive without using a p -value, and say why crossing is not required. (4) A reader concludes from the overlapping bands that the two slopes are the same. State what would have to be estimated instead.

Exercise 2: A conditional slope needs the covariance (Foundational)

In a fitted product-term model the focal coefficient is 2.6972 and the product coefficient is 0.1149. At centered moderator $w = 10$, the full covariance calculation gives a combined slope variance of 0.0379; the residual degrees of freedom are 316 and $t_{.975,316} = 1.9675$. Here $w = 10$ means ten units above the anchor. (1) Write the conditional slope and its three-term variance formula, retaining the covariance; calculate the slope at $w = 10$. (2) Use the supplied combined variance to calculate its SE and 95% interval. (3) A student omits the covariance term and reports 0.1865; another adds the focal standard error to ten times the product standard error and reports 0.2634. Say what each number is and why neither is the standard error required here. (4) To compare the slopes at $w = -10$ and $w = 10$, which quantity should be estimated directly? Explain why subtracting the endpoints of their two separate intervals does not supply an interval for that difference.

Exercise 3: Translate a centered model (Foundational)

A model fitted with the raw moderator has intercept 32.8571, focal coefficient -5.3448 , moderator coefficient 0.3822 and product coefficient 0.1149; the raw focal coefficient has $t = -5.5131$. The moderator is a test score and the analyst now centers it at 70. (1) Give the focal coefficient and the intercept of the centered model. (2) State which of the following change: the product coefficient, the fitted values, the residuals, the product-block $F = 68.3548$. (3) Explain what the raw focal coefficient's small p -value is a test of, and why that does not make the raw parameterization preferable. (4) A colleague says centering removed a confounder. State what centering did and did not do.

Exercise 4: A two-column product block (Foundational)

On the same 397 records, a model with a common service slope for three academic ranks has residual sum of squares 219014.23 on 393 degrees of freedom, and the model adding two service-by-rank product columns has 218471.17 on 391. The 5% critical value of $F(2, 391)$ is 3.0188. (1) Calculate the block F statistic. (2) State the decision at 5% and what it does and does not say about the three rank-specific slopes. (3) Explain why the result would be unchanged if the full professor rather than the assistant rank were the reference, while the two individual product t tests would not be. (4) Name one quantity a reader needs before accepting “the slopes are the same” as a summary.

Exercise 5: A difference of differences from cell means (Integrative)

In a conformity experiment the mean number of conforming responses is 8.9 with a low-status partner and 17.4 with a high-status partner among low-authoritarianism participants; 7.2500 and 14.2727 among medium-authoritarianism participants; and 12.6250 and 11.8571 among high-authoritarianism participants. (1) Give the partner-status difference in each authoritarianism category. (2) Give the difference of differences comparing the low and high categories. (3) Its standard error is 3.4507 on 39 degrees of freedom, with $t_{.975, 39} = 2.0227$; give the 95% interval and the t statistic. (4) The cell counts are 10, 5, 4, 11, 8 and 7 and the cell standard deviations range from 2.6437 to 7.3473. State one consequence for the reference distribution used in (3), and one thing the contrast does not establish about partner status.

Exercise 6: Report a sensitive conditional slope (Integrative)

A conditional slope is estimated as -0.3947 thousand dollars per year. Three standard errors are available for it: 0.2949 from the classical common-variance model, 0.1679 from an HC3 covariance, and 0.1736 from a 2,000-resample case bootstrap whose percentile interval is $[-0.8066, -0.1436]$. Use $t_{.975, 391} = 1.966$ for the first two. Residual standard deviations in the three groups are 8.0906, 13.2448 and 27.6842 against a pooled 23.6379. (1) Give the classical and HC3 intervals. (2) Say which intervals exclude zero and explain, using the residual standard deviations, why the calculations disagree. (3) Write one sentence reporting this slope for a reader, without choosing a winner. (4) A student joins pointwise bootstrap intervals into a curve and claims 95% simultaneous coverage. Explain the error and name one study problem that resampling does not repair.

Appendix A: Conditional quantities as linear combinations

One formula for four questions

A conditional slope, conditional mean, group contrast and interaction contrast can all be written $\mathbf{c}^\top \boldsymbol{\beta}$. The vector $\mathbf{c}$ names the question; the estimated variance is $\mathbf{c}^\top \widehat{\mathbf{V}} \mathbf{c}$. For coefficient order $(\beta_0, \beta_X, \beta_W, \beta_{XW})$:

Table 16

Different questions use the same coefficient covariance.

Quantity	Contrast vector $\mathbf{c}^\top$
Practice slope at w	$(0, 1, 0, w)$
Mean at (x, w)	$(1, x, w, xw)$
Mean difference $x_b - x_a$ at w	$(0, x_b - x_a, 0, w(x_b - x_a))$
Slope difference at w_b and w_a	$(0, 0, 0, w_b - w_a)$

At $w = 10$ the first row gives 3.8461 with SE 0.1947 from full precision. The faculty associate slope uses $\mathbf{c}^\top = (0, 1, 0, 0, 1, 0)$ and gives -0.3947 with classical SE 0.2949. The uncertainty belongs to the whole comparison, not a sum of separate coefficient SEs.

Centering translates the coefficients

For $W_c = W - c$,

$$\boldsymbol{\beta}^{\text{new}} = \mathbf{T}\boldsymbol{\beta}, \quad T = \begin{pmatrix} 1 & 0 & c & 0 \\ 0 & 1 & 0 & c \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{V}^{\text{new}} = \mathbf{T}\mathbf{V}\mathbf{T}^\top.$$

The last row fixes the product coefficient. The first two translate the anchor. With $c = 70$, the full-precision raw estimates give intercept 59.6123 and practice coefficient 2.6972. A translated contrast uses the translated covariance; its estimate and SE then agree with the original.

The boundary quadratic

Writing $a = \hat{\beta}_X$, $b = \hat{\beta}_{XW}$, $v_a = \widehat{\mathbf{V}}_{XX}$, $v_b = \widehat{\mathbf{V}}_{XW, XW}$ and $v_{ab} = \widehat{\mathbf{V}}_{X, XW}$, the squared endpoint condition becomes

$$Aw^2 + Bw + C = 0, \quad A = b^2 - t^2v_b, \quad B = 2ab - 2t^2v_{ab}, \quad C = a^2 - t^2v_a.$$

For distinct real roots, the band excludes zero outside them if $A > 0$ and inside if $A < 0$. With $A = 0$ the condition is linear; with no real roots the sign is constant and must be checked at a supported value. The generated fit has $A = 0.0125$ and baseline roots 39.039 and 51.284. Substitution into the quadratic gives numerical zero, and the pointwise interval touches zero there. The lower root remains outside the data.

Appendix B: What changes beyond this chapter

A three-way product XWZ allows a two-way pattern to vary with a third predictor. Hierarchy keeps all two-way products and all component terms. Its interpretation requires conditional comparisons across three dimensions; Chapter 4's three-way table gives the categorical reading.

With a nonlinear link, additivity on the linear-predictor scale and on the response scale are different properties. Even without a product, a logistic model can give probability differences that vary with another predictor. Always name the scale.

A conditional indirect effect concerns mediation. Its causal interpretation needs defensible temporal ordering, adequate functional form and assumptions about common causes of the intervening variable and outcome. An interval around an unidentified causal quantity leaves that problem in place.

Department, classroom or school dependence changes the appropriate uncertainty calculation. Adding a product changes a conditional mean; it does not make records independent.

Extended practice. (1) Refit the faculty model with log salary. Explain how the comparison changes from dollars to proportional changes, and why raw residual sums of squares or likelihood criteria on different outcome scales cannot be compared without accounting for the transformation. (2) Resample faculty within rank rather than across all records. Explain which sampling story fixes the rank counts and which lets them vary. Compare the associate slope intervals without treating the two schemes as independent substantive evidence.

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